Vertex Formula

1 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.
Complete the following statement:
The x-coordinate of the vertex of the parabola occurs at $x = \square$. The y-coordinate of the vertex is found by substituting $x = \square$ into the parabola's equation and evaluating the function at this value of x.

2 What does the line $x = \dfrac{-b}{2a}$ represent on the parabola defined by the equation $y = ax^2 + bx + c$ $(a \neq 0)$?

3 Consider the equation $y = 6x - x^2$.

a Find the x-intercepts of the quadratic function.

b Find the coordinates of the turning point.

4 Consider the parabola $y = x^2 - 18x + 10$.

a Is the graph of the function concave up or down?

b State the value of x at which the minimum value of the function occurs.

c State the minimum y-value of the function.

5 Consider the quadratic function $y = x^2 - 6x + 5$.

a Is the graph of the function concave up or down?

b Find the x-intercepts.

c State the the y-intercept.

d Determine the axis of symmetry.

e Hence or otherwise state the coordinates of the vertex of the parabola.

6 Consider the quadratic function $y = -x^2 - 2x + 8$.

a Is the graph of the function concave up or down?

b Find the x-intercepts of the function.

c Find the y-intercept.

d Determine the axis of symmetry.

e Hence or otherwise find the vertex of the curve.

7 Consider the quadratic function $y = -8 -$ [illegible] x^2.

a Is the graph of the function concave up or down?

b Find the x-intercepts of the function.

c Find the y-intercept.

d Determine the axis of symmetry.

e Hence or otherwise find the vertex of the curve.

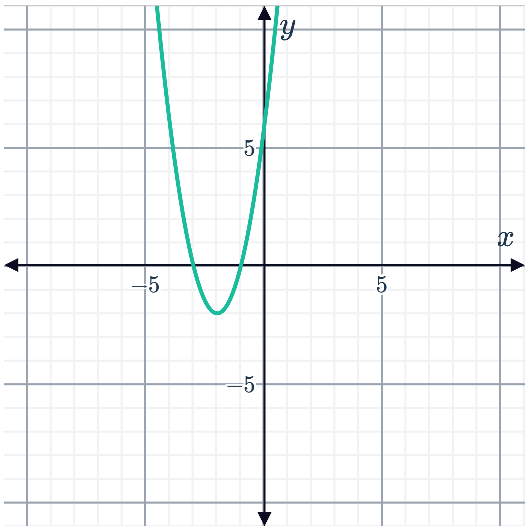
8 Consider the function $y = x^2 + 4x + 3$.

a Determine the equation of the axis of symmetry.

b Hence determine the minimum value of y.

c Hence state the coordinates of the vertex of the curve.

d Here is the graph of another quadratic function. State the coordinates of its vertex.



e Determine whether the following relationship is true between the graph of $y = x^2 + 4x + 3$ and the graph provided.

i They have the same x-intercepts.

ii They share the same turning point.

iii They have the same concavity.

9 Consider the function $y = (14 - x)(x - 6)$.

a State the zeros of the function.

b Find the axis of symmetry.

c Is the graph of the function concave up or concave down?

d Determine the maximum y-value of the function.

10 Find the coordinates of the vertex of $y = 3x^2 - 6x - 9$.

11 Find the maximum value of y for the quadratic function $y=-x^2+10x-25$.

12 Consider the quadratic function $f\left(x\right)=-7.5x^2-1.3x-1.3$.

a Find the x-coordinate of the vertex correct to three decimal places.

b Hence, find the maximum value obtained by the function, correct to two decimal places.

13 Find the x-coordinate of the vertex of the parabola represented by $P\left(x\right)=px^2-\dfrac{1}{2}px-q$.

14 Consider these two parabolas, labeled P_1 and P_2.
$P_1: y=x^2+4x+6$
$P_2: y=x^2-4x+6$

a Determine the coordinates of the vertex for each parabola.

b How far apart are the vertices of the two parabolas?

15 Consider the curve $y=x^2+6x+4$.

a Determine the axis of symmetry.

b Hence, determine the minimum value of y.

Applications

16 When an object is thrown into the air, its height above the ground is given by the equation

$$h=193+24s-s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

17 The height at time t of a ball thrown upwards is given by the equation $h=59+42t-7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the height of the ball at its highest point.